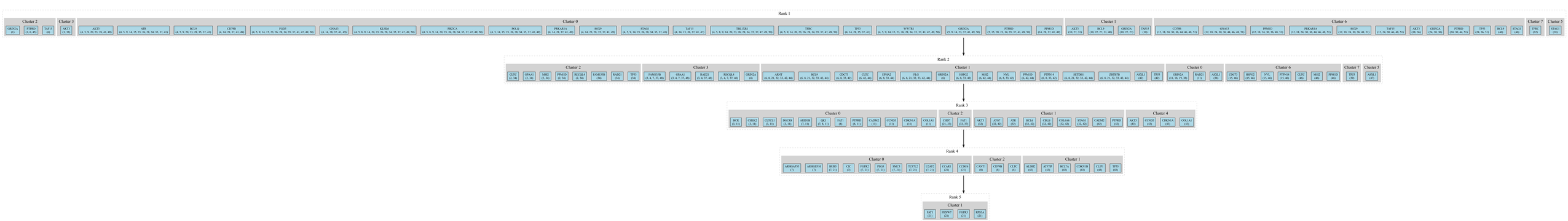

Rank 1
Cluster 2
GRIN2A (1)
PTPRD (1, 6, 45)
TAF15 (6)
Cluster 3
AKT3 (3, 33)
Cluster 0
AKT3 (4, 5, 9, 20, 23, 28, 41, 49)
ATR (4, 5, 9, 14, 15, 23, 26, 28, 34, 35, 37, 41)
BCL9 (4, 5, 9, 20, 23, 28, 35, 37, 41)
CD79B (4, 14, 28, 37, 41, 49)
FGD5 (4, 5, 9, 14, 15, 23, 26, 28, 34, 35, 37, 41, 47, 49, 50)
GNA13 (4, 14, 28, 37, 41, 49)
KLHL6 (4, 5, 8, 9, 14, 20, 23, 26, 28, 34, 35, 37, 47, 49, 50)
PIK3CA (4, 5, 8, 9, 14, 20, 23, 26, 28, 34, 35, 37, 47, 49, 50)
POLQ (4, 5, 14, 15, 23, 26, 28, 34, 35, 37, 41, 49)
PRKAR1A (4, 14, 28, 37, 41, 49)
SOX9 (4, 14, 23, 28, 35, 37, 41, 49)
STAG1 (4, 5, 9, 14, 23, 26, 28, 34, 35, 37, 41)
TAF15 (4, 14, 15, 26, 37, 41, 47)
TBL1XR1 (4, 5, 8, 9, 14, 20, 23, 26, 28, 34, 35, 37, 47, 49, 50)
TFRC (4, 5, 9, 14, 20, 23, 26, 28, 34, 35, 37, 47, 49, 50)
TP53 (4, 14, 28, 35, 37, 41)
WWTR1 (4, 5, 9, 14, 15, 23, 26, 28, 34, 35, 37, 41, 47, 49, 50)
GRIN2A (5, 9, 14, 23, 37, 41, 49, 50)
PTPRD (5, 15, 20, 23, 34, 35, 37, 41, 49, 50)
PPM1D (14, 28, 37, 41, 49)
Cluster 1
AKT3 (10, 27, 31)
BCL9 (10, 22, 27, 31, 40)
GRIN2A (10, 22, 27)
TAF15 (10)
Cluster 6
CD79B (12, 18, 24, 30, 36, 44, 46, 48, 51)
GNA13 (12, 18, 24, 30, 36, 44, 46, 48, 51)
PPM1D (12, 18, 24, 30, 36, 48, 51)
PRKAR1A (12, 18, 24, 30, 36, 44, 46, 48, 51)
SOX9 (12, 18, 24, 30, 36, 48, 51)
TAF15 (12, 24, 30, 46, 48, 51)
AKT3 (18, 36)
GRIN2A (24, 30, 36)
PTPRD (24, 30, 44, 51)
TP53 (24, 36, 51)
BCL9 (46)
STAG1 (46)
Cluster 7
TFRC (32)
Cluster 5
STAG1 (38)
Rank 2
Cluster 2
CLTC (2, 34)
GPAA1 (2, 34)
MSI2 (2, 34)
PPM1D (2, 34)
RECQL4 (2, 34)
FAM135B (34)
RAD21 (34)
TP53 (34)
Cluster 3
FAM135B (3, 4, 7, 37, 48)
GPAA1 (3, 4, 7, 37, 48)
RAD21 (3, 4, 37, 48)
RECQL4 (3, 4, 7, 37, 48)
GRIN2A (4)
Cluster 1
ARNT (6, 8, 21, 32, 33, 42, 44)
BCL9 (6, 8, 21, 32, 33, 42, 44)
CDC73 (6, 8, 33, 42)
CLTC (6, 42, 44)
EPHA2 (6, 8, 33, 44)
FLG (6, 8, 21, 32, 33, 42, 44)
GRIN2A (6)
HSPG2 (6, 8, 33, 42)
MSI2 (6, 42, 44)
NVL (6, 8, 33, 42)
PPM1D (6, 42, 44)
PTPN14 (6, 8, 33, 42)
SETDB1 (6, 8, 21, 32, 33, 42, 44)
ZBTB7B (6, 8, 21, 32, 33, 42, 44)
ASXL1 (42)
TP53 (42)
Cluster 0
GRIN2A (11, 18, 19, 38)
RAD21 (11)
ASXL1 (38)
Cluster 6
CDC73 (15, 46)
HSPG2 (15, 46)
NVL (15, 46)
PTPN14 (15, 46)
CLTC (46)
MSI2 (46)
PPM1D (46)
Cluster 7
TP53 (39)
Cluster 5
ASXL1 (47)
Rank 3
Cluster 0
BCR (3, 11)
CHEK2 (3, 11)
CLTCL1 (3, 11)
DGCR8 (3, 11)
ARID1B (7, 11)
QKI (7, 8, 11)
FAT1 (8)
PTPRD (8, 11)
CADM2 (11)
CCND3 (11)
CDKN1A (11)
COL1A1 (11)
Cluster 2
CHD7 (21, 33)
FAT1 (33, 37)
Cluster 1
AKT3 (32)
ATG7 (32, 42)
ATR (32)
BCL6 (32, 42)
CBLB (32, 42)
COL6A6 (32, 42)
STAG1 (32, 42)
CADM2 (42)
PTPRD (42)
Cluster 4
AKT3 (43)
CCND3 (43)
CDKN1A (43)
COL1A1 (43)
Rank 4
Cluster 0
ARHGAP35 (7)
ARHGEF10 (7)
BUB3 (7, 21)
CIC (7)
FGFR2 (7, 21)
PEG3 (7, 21)
SMC3 (7, 21)
TCF7L2 (7, 21)
U2AF2 (7, 21)
CCAR1 (21)
CCDC6 (21)
Cluster 2
CANT1 (8)
CD79B (8)
CLTC (8)
Cluster 1
ALDH2 (43)
ATF7IP (43)
BCL7A (43)
CDKN1B (43)
CLIP1 (43)
TP53 (43)
Rank 5
Cluster 1
FAT1 (21)
FBXW7 (21)
FGFR3 (21)
RPS3A (21)